Adamya Kumar

AI/ML Engineer | Generative AI Engineer | LLM Applications

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TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript

GenAI and LLMs: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LangSmith, Prompt Engineering, Embeddings, OpenAI APIs, Google Gemini, Groq, Hugging Face

Agentic Systems: AI Agents, Multi-Agent Architecture, Tool Calling, Memory Management, State Management

Vector Databases: ChromaDB, FAISS, Pinecone, Semantic Search

Machine Learning: Scikit-learn, Pandas, NumPy, Neural Networks

Deep Learning: TensorFlow, Natural Language Processing (NLP), Transformer Architectures

Backend and Deployment: FastAPI, REST APIs, Docker, AWS, CI/CD, Streamlit, Render, Vercel, Hugging Face Spaces

Databases: PostgreSQL, MySQL, MongoDB, SQLite

AI Development Tools: GitHub Copilot, Claude, Cursor, Spec-Driven Development

Core: Data Structures and Algorithms, Problem Solving, Testing, Git, GitHub, Jupyter

PROJECTS

Production RAG Bot | *Python*, *LangChain*, *ChromaDB*, *Groq*, *Google Gemini*, *Hugging Face*, *Streamlit*
Live: pro-rag-chatbot.streamlit.app

- Built a Retrieval-Augmented Generation (RAG) chatbot for document Q&A where users upload multiple PDFs; handles PDF parsing (pypdf), chunking, embeddings, and retrieval-based answers with LangChain.
- Supported Groq (Llama 3.1 and Llama 3 70B) and Google Gemini (2.0 and 2.5 Flash) with provider-specific embeddings (sentence-transformers for Groq, Google embeddings for Gemini), so users can switch models without starting over.
- Persisted vectors in ChromaDB with automatic reuse and append-on-upload, so documents are not re-processed.
- Added a developer view to inspect the stored chunks behind each answer, plus chat history export and clear/reset/undo controls; deployed on Streamlit Cloud with session-based chat state.

ResearchMind, Multi-Agent AI Research System | *Python*, *LangChain*, *Streamlit*, *Docker*, *Tavily*, *OpenRouter*
Live: researchmind-multi-agent-ai-research.onrender.com

- Built an agentic research workflow with four stages: a Search agent (Tavily) finds recent sources, a Reader agent scrapes them with BeautifulSoup, a Writer chain drafts the report, and a Critic chain reviews and scores it.
- Created a Streamlit interface that shows each agent stage in real time and lets users download the final report as Markdown.
- Containerized with Docker and deployed on Render; kept API keys in environment variables with an .env.example template.

Spark Commerce, AI Shopping Assistant | *FastAPI*, *LangGraph*, *FAISS*, *Gemini*, *Express.js*, *React*, *MongoDB*
Live: www.carteer.space

- Designed the LLM-powered shopping assistant as a separate FastAPI and LangGraph service with tools for product search, cart, and orders, using FAISS retrieval and Gemini; the Express API proxies chat over SSE and the assistant calls the store API with the user token. Admin product changes are planned to sync into FAISS and SQLite metadata.
- Built the full-stack store it runs on (React, Express, MongoDB, JWT cookie auth, Cloudinary) and deployed it on Vercel with a custom domain; used spec-driven development with Cursor and GitHub Copilot.

Coffee Roast Quality Predictor | *Python*, *Scikit-learn*, *Streamlit*, *Hugging Face Spaces*
Live: huggingface.co/spaces/Adamya2307/coffee-roast-predictor

- Trained a Multi-Layer Perceptron (ReLU, Adam, L2 regularization) to predict coffee roast quality from bean characteristics, reaching 89.8% accuracy on a held-out test set.
- Built and deployed an interactive Streamlit interface on Hugging Face Spaces.

EDUCATION

Bachelor of Technology in Computer Science and Engineering Sep. 2021 – Jun. 2025
Graphic Era Hill University, Dehradun, India

CERTIFICATIONS AND ACHIEVEMENTS

- Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms (Coursera, DeepLearning.AI).
- Top 10 Contributor, Hacktoberfest 2023 (among 2,000+ developers); solved 250+ DSA problems on LeetCode and Codeforces.